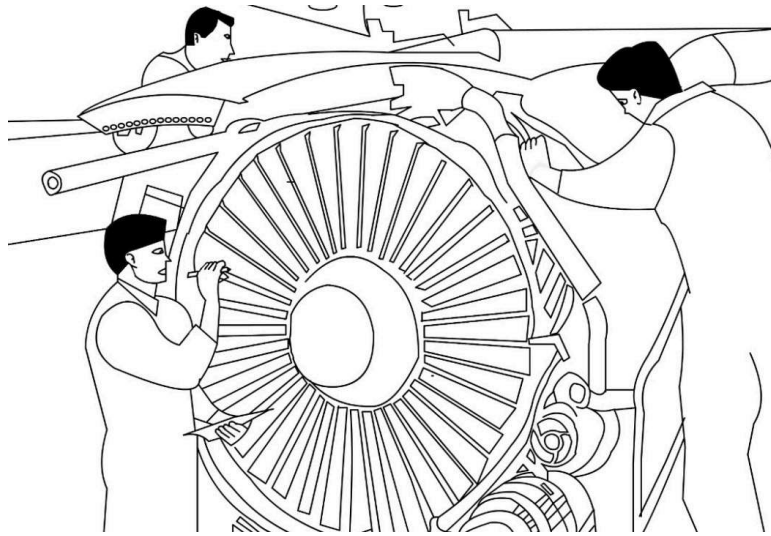




Gas Turbine Engine Academic Courses and International Standards Mapping



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My view is based on my experience working on industry gas turbines by Solar Turbines Centaur Series at Airod Sdn Bhd. My work was developing the repair procedures of those engines under maintenance. I was also responsible to rectify any discrepancies of the engine after the test run in the test cell. In addition to the industrial gas turbines, I also had the experience in aero gas turbine engine; Allison T56 installed on the C130, Turbomeca Artouse installed on Alouette III helicopter, General Electric J85 engines installed on F5E, General Electric CT58-140-1 on the Nuri S61N, and modern engine Pratt & Whitney Canada PT6 on several general aviation aircraft. I also had done type validation of the CFM56-5B installed on the A320 family, Turbomeca TM 333-2B2 turboshaft engine installed on the Indian DHRUV ALH Helicopter and Klimov TV3-117VM turboshaft engines installed on the Mi-171 helicopter. On engine testing, I also had experience working on water dynamometer cavitation problems. The review is on academic program of engine gas turbine course, which is

management of gas turbine assets. It excluded engine design and manufacturing.

Gas Turbine Engine Types

Gas turbine engines have been used in many applications, transportation, ship, aeroplane, power generation, pumps, and gas compressor. For aircraft applications the design requirements are in accordance with ICAO Annex 8. However, some of the aircraft design and certified gas turbine engine is used for industrial (other than aircraft used) application; this engine is called the Aero-Derivative Gas Turbine Engine. Aero derivative land turbines produce smaller power but are more compact, run at higher efficiency and pressure ratios. It emerged in 1960. The other category is gas turbine design and manufacture for non-aeronautical application, where the engines are heavier and closely integrated to the devices they are powered, it is often an electric generator. The size ranges from portable mobile plants to large, complex systems weighing more than a hundred tones housed in purpose-built buildings.

Mapping Subjects Competency Standards

The students at bachelor course in gas turbine shall be exposed to the gas turbine engine theories. For industrial gas turbines the students are required to understand the industrial standards of a gas turbine to ensure the students employability.

The subjects for the course should align with the overall competency requirement by the industry. The gas turbine engineer works in the industry to manage the **repairs and performs maintenance on the mechanical apparatuses of gas turbine engines**. They inspect the engines and components, work with blueprints and charts, and operate testing equipment or they may work in engine rooms or shops, often on ships. They are graduates in a Mechanical Engineering discipline. They shall have knowledge on engineering, testing, commissioning, operation & maintenance of turbo-machinery Equipment. They also shall have knowledge and skills in condition monitoring, troubleshooting, root-cause analysis and computers and database management. Other attributes required are sound techno-commercial acumen, contract management skills, strong and dynamic leadership for effective technical and business contributions for interaction with company management. Some companies require fully

Office), good communicator and organizer self-motivated with excellent supervisory and team leadership skills, good knowledge and understanding of the Oil and Gas process and systems and should be aware of Health Safety and Environment and process safety requirements. The ISO has seriously advocating the introduction of standards in universities subjects. Refer to <https://www.iso.org/education-about-standards.html#WSC>

Competencies

The competencies for the program shall be clear and focus on the main potential employers, which are gas turbine operations and maintenance. The engine design is normally done by mechanical design and electrical design engineering background. However, the students are encouraged to understand the background of the gas turbine engine design by introducing the standards. In the industry, every design, services, and activities are linked to standards as the guidance and these standards become very important when certifications are required, for example the personnel certification ***"ENGINEER'S CERTIFICATE OF COMPETENCY (STEAM AND INTERNAL COMBUSTION ENGINES"***. This competency certificate is required for the operator of a gas turbine.

Industrial Gas Turbine Standards

There are several related standards for the gas turbine design, certification, testing and associated requirements such as the environment and Occupational Health and Safety.

1. **ISO 19859:2016 Gas turbine applications — Requirements for power generation.** It is not applicable for gas turbines used for all types of propulsion including aircraft, mobile barges, floating production vessels and marine propulsion applications and microturbines
2. **ISO 11086:1996 Gas turbines — Vocabulary** - The definitions of this International Standard are presented in groupings for ease of use. See contents. [Annex A](#) presents pictorial examples of gas turbine systems which are specified in [ISO 3977](#). [Annex B](#) also presents pictorial examples of combined cycles. The intent of the annexes is to clarify the meaning of the definitions.

for ease of reference. The format of the definitions has been altered to achieve a uniformity of definition presentation. The definition of gas turbines in [ISO 2314](#) and [ISO 3977](#) is general in nature and suitable for the procurement and acceptance standards, whilst the definition of [1.1](#), gas turbine engine, is for a single unit.

3. **ISO 3977 is an international standard related to the design and procurement** of gas turbine system applications. ISO 3977 is based primarily on the ASME 133 series on gas turbines, as well as the API 616 and API 11PGT standards. The standard environmental design point of any gas turbine system is 15 °C, 60% relative humidity, and sea level elevation. The standard is divided into eight parts and covers procurement, design requirements, installation, and reliability.
4. **ISO 3977-1:1997 Part 1: General introduction and definitions.** It provides technical information to be used for the procurement of gas turbine systems, including combined-cycle systems and their auxiliaries, by a purchaser from a manufacturer. Where the term "manufacturer" is used in this International Standard, it is deemed to mean the gas turbine manufacturer or the appropriate responsible contractor. This International Standard provides a basis for the submission of proposals in line with the different environmental and safety requirements. It also specifies, wherever possible, criteria to establish whether these are met. It does not attempt to deal with local or national legal requirements to which the installation may be required to conform. Because of the very widely varying operating modes for gas turbines in practice, distinct categories of operating modes are specified with which a "standard" rating can be associated. These ratings are made based on the ISO standard ambient reference conditions. The various parts of [ISO 3977](#) define a standard framework for dealing with questions of fuel and other matters, such as the minimum information to be provided by both the purchaser and the manufacturer. They do not, however, purport to include all necessary information for a contract and each gas turbine installation should be considered in its entirety. Attention is drawn to the need for technical consultation between the manufacturer and the purchaser to ensure compatibility of equipment being supplied, particularly where the responsibility for supply is divided. [ISO 3977](#) is applicable to open-cycle gas turbine power plant

closed-cycle and combined-cycle gas turbine power plants. In the case of turbines using free piston gas generators or special heat sources (e.g., chemical process, nuclear reactors, furnaces for supercharged boilers), it may be used as a basis but will need to be suitably modified. This International Standard is not applicable to gas turbines used to propel aircraft, road construction and earth-moving machines, agricultural and industrial types of tractors and road vehicles.

5. **ISO 3977-2:1997 Part 2: Standard reference conditions and ratings.** The following standard contains provisions which, through reference in this text, constitute provisions of this part of [ISO 3977](#). At the time of publication, the edition indicated was valid. All standards are subject to revision, and parties to agreements based on this part of [ISO 3977](#) are encouraged to investigate the possibility of applying the most recent edition of the standard indicated below. Members of IEC and ISO maintain registers of currently valid International Standards. [ISO 2314:1989](#), *Gas turbines — Acceptance tests*.
6. **ISO 3977-3:2004 Part 3: Design requirements.** ISO 3977-3:2004 covers the design requirements for the procurement of all applications of gas turbines and gas turbine systems, including gas turbines for combined cycle systems and their auxiliaries, by a purchaser from a packager. It also provides assistance and technical information to be used in the procurement. It is not intended to deal with local or national legislative requirements with which the installation may be required to conform. ISO 3977-3:2004 is applicable to simple-cycle, combined-cycle and regenerative-cycle gas turbines working in open systems. It is not applicable to gas turbines used to propel aircraft, road construction and earth moving machines, agricultural and industrial types of tractors and road vehicles. In cases of gas turbines using special heat sources (for example, chemical process, nuclear reactors, furnace for a supercharged boiler), ISO 3977-3:2004 provides a basis.
7. **ISO 3977-4:2002 Part 4: Fuels and environment.** This part of ISO 3977 provides guidelines for procurement of gas turbines with consideration of the fuel quality and of the environmental performance. Guidance is given to both the packager and purchaser on what information should be provided with regard to the fuel used by a gas

impact. Fuel specifications are referenced but not provided.

8. **ISO 3977-5:2001 Part 5: Applications for petroleum and natural gas industries.** This part of [ISO 3977](#) specifies requirements and gives recommendations for the design, materials, fabrication, inspection, testing and preparation for shipment of packaged gas turbines for use in drilling, production, refining and the transport by pipelines of petroleum and natural gas. It is applicable to the procurement of gas turbines and gas turbine systems, including gas turbines for combined cycle systems, and their auxiliaries by a purchaser from a packager. This part of [ISO 3977](#) is not intended to deal with local or national legislative requirements to which the installation may be required to conform.
9. **ISO 3977-6: 2000 Superseded By: ISO DIS 3977-6** February 10, 2000
10. **ISO 3977-7:2002 Part 7: Technical information.** This part of ISO 3977 specifies the information that needs to be submitted during the proposal and contract stages of a project for the entire scope of supply for which the packager will assume technical and contractual responsibility.
11. **ISO 3977-8:2002 Part 8: Inspection, testing, installation, and commissioning.** This part of ISO 3977 states the principles for systems and procedures to assure the integrity of a packager's product and services. It gives guidance on the inspection, testing, installation and commissioning required for the package and packaged equipment. It outlines the responsibilities between the purchaser and packager for inspection, coordination, reporting and recording.
12. **ISO 3977-9:1999 Part 9: Reliability, availability, maintainability, and safety.** The purpose of this part of [ISO 3977](#) is to provide a basis for exchange of information about reliability, availability, maintainability and safety between gas turbine manufacturers, users, consultants, regulatory bodies, insurance companies and others. It defines terms and definitions used within this part of [ISO 3977](#) and also describes component life expectancy, repairs and criteria for determining overhaul intervals. This part of [ISO 3977](#) is applicable to all elements of the gas turbine, especially, but not limited to, the following: — compressor, — turbine, — combustion system, — intercooler, — regenerator or recuperator, — air ducting system, — exhaust

system, — fuel system, — lubrication system,
— cooling water system, — rotor bearings, — gears,
— coupling, — starting equipment,
— baseplate/foundation, — enclosures and
ventilation system.

13. **ISO 10494:2018 Turbines and turbine sets — Measurement of emitted airborne noise — Engineering/survey method.** ISO 10494:2018 specifies methods for measuring the noise emission of a turbine or turbine set under steady-state operating conditions. It specifies methods for measuring the sound pressure levels on a measurement surface enveloping a source, and for calculating the sound power level produced by the source
14. **ISO 11042-1:1996 Gas turbines — Exhaust gas emission — Part 1: Measurement and evaluation.** Contains the methods used for the measurement and evaluation of the emission of the exhaust gases from gas turbines and defines appropriate emission terms. Presents requirements for the test environment and instrumentation as well as the accuracy of measurement and correction of data. Does not apply to gas turbines used in aircraft.
15. **ISO 11042-2:1996 Gas turbines — Exhaust gas emission — Part 2: Automated emission monitoring.** Establishes the monitoring programme and the requirements for the selection and operation of hardware to be used for continuous gas emission measurements. Monitors the concentration and absolute magnitude of specified emissions in the exhaust gas as well as related gaseous components. Requires that the following parameters be continuously monitored: emissions, diluent gases (oxygen, carbon dioxide), exhaust gas flow, fuel consumption and gas turbine performance.
16. **ISO 18888:2017 Gas turbine combined cycle power plants — Thermal performance tests.** Specifies standard rules for preparing, conducting, evaluating, and reporting thermal performance tests on combined cycle and cogeneration power plants driven by gas turbines for base and part load operation with or without supplementary firing.
17. **ISO 19372:2015 Microturbines applications — Safety.** Covers the safety requirements for industrial, commercial, and residential automated stationary microturbine (micro gas turbine) engine generator assemblies with an individual system or subsystem output rating with nominal supply voltages not

not exceeding 1 500 V for direct current (D.C.), 500 kW per individual unit/subsystem or less that are intended for installation and use in ordinary and hazardous location using liquid or gaseous fuels and the safety related control and detection systems and essential auxiliaries for all types of open cycles, closed-cycle, and semi-closed (simple, combined, regenerative, reheat, etc.) used in onshore and offshore applications including floating production platforms.

18. **ISO 19859:2016 Gas turbine applications — Requirements for power generation.** It specifies the minimum technical and documentation requirements for the evaluation and procurement of gas turbine systems for electrical power generation. It applies to simple cycle and combined cycle gas turbines for both onshore and offshore applications, where applicable. It also applies to gas turbines used in cogeneration (see ISO 11086:1996, Annex B). Testing of the gas turbine in combination with a generator is included in the scope.
19. **ISO 19860:2005 Gas turbines — Data acquisition and trend monitoring system requirements for gas turbine installations.** Applies to data-acquisition and trend-monitoring systems for gas turbine installations and associated systems. ISO 19860:2005 classifies and defines monitoring systems and their technical terms and establishes a system for conversion and validation of measured quantities to enable a comparison of the various systems, their features and their performances.
20. **ISO 21789:2009 Gas turbine applications — Safety.** Covers the safety requirements for gas turbine applications using liquid or gaseous fuels and the safety-related control and detection systems and essential auxiliaries for all types of open cycles (simple, combined, regenerative, reheat, etc.)
21. **ISO 21905:2020 Gas turbine exhaust systems with or without waste heat recovery.** This document specifies requirements and gives recommendations for the design, materials of construction, modelling, controlling, fabrication, inspection, testing, installation, start-up and operation of industrial gas turbine (GT) exhaust systems with or without waste heat recovery unit (WHRU). Gas turbines can be on-shore or off-shore for such sectors as oil and gas, chemical and process industries, utilities, or other intensive energy users. For this document, the exhaust system means all items in the turbine

atmosphere. The following items are not covered by this document:— heat recovery steam generator equipment (HRSG); — supplementary fired systems — auxiliary fire systems.— exhaust gas collector (also known as exhaust plenum);— fire detection and extinguishing systems.— emissions control equipment intended to modify the gaseous composition of the exhaust gas.— WHRUs that are of the firetube type, where the turbine exhaust gas (TEG) passes through the tubes.

22. **ISO 26382:2010 Cogeneration systems — Technical declarations for planning, evaluation, and procurement.** ISO 26382:2010 describes the technical declarations for a cogeneration system (CGS) that simultaneously supplies electric power and heating and/or cooling, for planning, evaluation, and procurement. It applies to the identification of investigation items for project evaluation, CGS evaluation, and primary information works for CGS procurement. It also specifies necessary check items in CGS planning, provides a procedure to obtain the satisfactory configuration of the CGS for each project, and includes a detailed process diagram of the key development steps.
23. **ISO 2314 Third edition 2009-12-15 Gas turbines — Acceptance tests** were developed based on ISO/TC 192 Gas turbines. Standardization in the field of all aspects of gas turbine design, application, installation, operation, and maintenance, including simple turbine cycles, combined cycle systems, definitions, procurement, acceptance, performance, environment (on the gas turbine itself and the external environment) and methods of test. ISO / TC 192 is responsible for preparing horizontal standards for all types of gas turbines. Work on aero gas turbine engines shall be undertaken in liaison with those technique committees having the primary responsibility. Note: ISO / TC 20 has the primary responsibility of preparing standards relative to the specific application of gas turbines to aerospace.

Aero derivative Gas Turbine Design Standards (CAAM CAD 8 - Airworthiness of Aircraft) - Part VI

Engines (Design and Certification). The ICAO developed the standards and recommended practice for design of an aircraft gas turbine engine in Annex 8, Part VI. These design standards are for the core engine, the engine accessories and control system are designed and certified as part of the aircraft. The following are the design

1 General

1.1 Applicability

1.1.1 Except as noted below, the Standards of this part are applicable to engines of all types, used as primary propulsion units, as required in Parts IIIB, IVB and V. The Standards of this part are applicable to an engine type at the time of submission of an application to CAAM for a type approval.

1.1.2 The level of airworthiness defined by the appropriate parts of the comprehensive and detailed national code for the engines designated in 1.1.1 shall be at least substantially equivalent to the overall level intended by the broad Standards of this part.

1.2 Engine installation and interfaces

1.2.1 All necessary information for the safe and correct interfaces between the engine and the aircraft shall be made available.

1.2.2 The installation instructions shall specify those assumptions concerning the conditions that may be imposed on the engine when it is eventually installed in an aircraft.

1.3 Declared ratings, conditions and limitations

1.3.1 The thrust or power ratings and the conditions of the atmosphere upon which they are based and all operating conditions and limitations which are intended to govern the operation of the engine shall be declared.

1.3.2 Within the stated limits of 1.3.1, the engine shall produce the thrust or power demanded of it at all required flight conditions, taking into account environmental effects and conditions.

1.4 Continuing airworthiness – maintenance information

1.4.1 General

Information for use in developing procedures for

be made available. The information shall include that described in 1.4.2, 1.4.3 and 1.4.4.

1.4.2 Maintenance information

Maintenance information shall include a description of the engine and recommended methods for the accomplishment of maintenance tasks. Such information shall include guidance on defect diagnosis.

1.4.3 Maintenance programme information

Maintenance programme information Maintenance programme information shall include the maintenance tasks and the recommended intervals at which these tasks are to be performed.

1.4.4 Mandatory maintenance requirements resulting from the type design approval

Mandatory maintenance requirements that have been specified by the State of Design as part of the approval of the type design shall be identified as such and included in the maintenance information of 1.4.3.

2 Design and Construction

2.1 Functioning

The engine shall be designed and constructed so as to function reliably within its operating limitations under its anticipated operating conditions when installed in accordance with Parts IIIB, IVB and V of this CAD and, if applicable, fitted with a propeller approved for the installation.

2.2 Failure analysis

For turbine engines, a safety assessment of the engine shall be conducted to ensure that it functions safely throughout the full range of operating conditions.

A summary shall be made of all foreseeable failures and combinations of failures that result in hazardous

If the primary failure of single elements (for example, disks) is likely to result in hazardous engine effects, reliance shall be placed on meeting prescribed integrity requirements.

2.3 Materials and manufacturing methods

The selection of materials and the manufacturing methods and processes shall account for the operational environment of the engine expected in service.

The materials and manufacturing methods and processes used in the construction of the engine shall result in known and reproducible structural behaviour.

2.4 Integrity

The integrity of the engine shall be demonstrated throughout its operating envelope and be maintained for its operational life.

The effects of cyclic loading, environmental and operational degradation and likely subsequent part failures shall not reduce the integrity of the engine below acceptable levels.

All necessary instructions for ensuring continued airworthiness in this regard shall be promulgated.

3 TESTS

The engine type shall complete satisfactorily such tests as are necessary to verify the validity of the declared ratings, conditions and limitations and to ensure that it will operate satisfactorily and reliably.

The tests shall include at least the following:

- a) **Power calibration.** Tests shall be conducted to establish the power or thrust characteristics of the engine when new and also after the tests in b) and c).
- c). There shall be no excessive decrease in power at the conclusion of all the tests specified.

and other characteristics are satisfactory and to demonstrate adequate margins of freedom from detonation, surge, flutter or other detrimental conditions as may be appropriate to the particular type engine.

c) **Endurance.** Tests of sufficient duration shall be conducted at such powers, thrust, speeds, temperatures and other operating conditions as are necessary to demonstrate reliability and durability of the engine. They shall also include operation under conditions in excess of the declared limits to the extent that such limitations might be exceeded in actual service.

d) **Operating Environment.** Tests shall be conducted to ensure that the engine characteristics are satisfactory with regard to the operating environment.

These aircraft engines may also later be used for power generation such as the GE. GE is taking the world's largest jet engine and turning it into a power plant. The machine's beating heart comes from the GE90-115B, which is the largest and most powerful jet engine, capable of producing 127,900 pounds of thrust, according to Guinness World Records. The electricity generator, which GE calls LM9000, will be able to generate a whopping 65 megawatts — enough to supply 6,500 homes — and reach full power in 10 minutes. "We picked the best technology across GE and built the biggest and most powerful aero derivative engine ever made," says GE Oil & Gas' Maurizio Ciofini, the engineering director of the project.

Based on the technology the basic core engine technology is similar, the differences are just the aircraft, once the engine is brought in for power generation, the accessories and control are in accordance with the needs to generate electricity.

Mapping of Syllabus and Standards Competency

The courses develop for employment by the industry. Therefore, each subject in course must be aligned towards industry needs and competency. The above standards are mapped against the subjects. Some of the examples as follows;

safety. The possible standard is the ISO/WD 24495-1
Plain language — Part 1: Governing principles and
guidelines

2. **Managing Effective Project** . The students are required to understand the project management cycles and the possible standard is the ISO 21500:2021 - Project, programme and portfolio management — Context and concepts
3. **Engineering Ethics** . The students are expected to be competent on BEM and MBOT Codes of Ethics. The subject may refer to the BOARD OF ENGINEERS MALAYSIA CIRCULAR NO. 001 CODE OF CONDUCT OF REGISTERED PERSON
4. **Internet of Things (IoT)**. The students are expected to be competent of the IoT standard ISO/IEC 21823-1:2019 - Internet of things (IoT) — Interoperability for IoT systems — Part 1: Framework ISO/IEC 20924:2021(MAIN) Internet of Things (IoT) - Vocabulary
5. **Applied Engineering Statistics**. The students are expected to be competent in Inferential statistics SPSS/SmartPLS for predictive trend study
6. **Computer Programming** . The students are expected to competent in computer programming focus on data mining for Data Analytics (i.e, Python, etc)
7. **Innovation Management**. The students are expected to be competent Innovation is always in the organization. ISO 56002:2019 Innovation management — Innovation management system — Guidance
8. **Turbomachinery**, the students are expected to understand the gas turbine engine application other than power generation. The ISO 26382:2010 Cogeneration systems — Technical declarations for planning, evaluation and procurement is one of the aspects to be taught.
9. **Computer Aided Design**. The STANDARDS BY ISO/TC 10/SC 6 Mechanical engineering documentation is relevant as part of the subject.
10. **CAM , CNC and Smart Manufacturing Technologies**. ISO/TS 14649-201:2011Industrial automation systems and integration — Physical device control — Data model for computerized numerical controllers — Part 201: Machine tool data for cutting processes
11. **Aviation Gas Turbine**. EASA CS-E Engines14 CFR Part



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12. **Industrial Gas Turbine.** ISO 3977-3:2004 Gas turbines — Procurement — Part 3: Design requirements
13. **Engine Fuel, Ignition and Start Systems.** ISO 3977-4:2002 Gas turbines — Procurement — Part 4: Fuels and environment
14. **Quality Assurance & Reliability.** The ISO9001/AS9110/AS9120/ISO 14224:2016(en) Petroleum, petrochemical and natural gas industries — Collection and exchange of reliability and maintenance data for equipment and ISO 3977-9:1999 Gas turbines — Procurement — Part 9: Reliability, availability, maintainability and safety are applicable.
15. **Environmental Impact of Engine Noise and Emission.** The students are expected to understanding on regulations DOE and ISO 14001 Standard: Environmental Management System is applicable.
16. **Integrated Occupational Safety, Health and Environment,** the students are required to understand the regulations of OSH and DOE. The ISO 45001 Standard: Occupational Health and Safety Management System is applicable as part of the subject.
17. **Data Analytics,** data mining and analysis for descriptive (diagnose), predictive and prescriptive. ISO 19731:2017 Digital analytics and web analyses for purposes of market, opinion and social research — Vocabulary and service requirements
18. **FMEA,** Risk assessment for installation, operation and maintenance, standard applied is the IEC60812 Analysis techniques for system reliability – Procedure for failure mode and effects analysis (FMEA)

Conclusion

The above approach will be innovation in developing courses, syllabus and subjects. Each SME should review each subject to related standards, regulations based on the competency required for the subjects. The above contents were taken from various standards.



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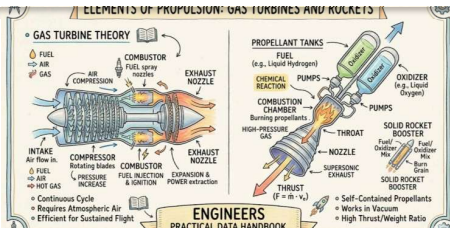
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